

1 Proving Q

In order to compute the risk-neutral probability q , we need to start with the Replicated riskless Payoff H , which has value $V_1 = H_{\text{up}}$ or H_{down} depending on the states. Then, we want to maximise the expected utility of the replicated portfolio. This will allow us to derive the results and find the risk-neutral probability q , which makes each states indifferent.

$$V_1 = \alpha S_1 + \beta \cdot (1 + r) = \begin{cases} \alpha \cdot S_0 u + \beta B_0 = X_{\text{up}} \\ \alpha \cdot S_0 d + \beta B_0 = X_{\text{down}} \end{cases}$$

We want to maximise $E[u(v_1)]$:

$$\max_{\alpha, \beta} E[u(\alpha_i S_1 + \beta_i R)] \quad \text{s.t.} \quad \alpha_i S_0 + \beta_i B_0 = V_0$$

We maximise the Expected Utility using the Lagrangian Function:

$$\begin{aligned} \mathcal{L}(\alpha, \beta, \lambda) &= E[u(\alpha_i S_1 + \beta_i B_1)] - \lambda(\alpha_i S_0 + \beta_i B_0 - V_0) \\ &= p \cdot u(X_u) + (1 - p) \cdot u(X_d) - \lambda(\alpha_i S_0 + \beta_i B_0 - V_0) \\ &= p \cdot u(\alpha S_u + \beta B_0) + (1 - p) \cdot u(\alpha S_d + \beta B_0) - \lambda(\alpha_i S_0 + \beta B_0 - V_0) \end{aligned}$$

$$\begin{aligned} \mathcal{L}_\alpha &= p \cdot u'(X_u) \cdot S_u + (1 - p) \cdot u'(X_d) \cdot S_d - \lambda \cdot S_0 = 0 \\ &= E[u'(V_1) \cdot S_1] = \lambda S_0 \end{aligned} \tag{I}$$

$$\longrightarrow \quad \lambda = \frac{E[u'(V_1) \cdot S_1]}{S_0}$$

$$S_0 = \frac{1}{\lambda} \cdot E[u'(V_1) \cdot S_1]$$

$$\begin{aligned} \mathcal{L}_\beta &= p \cdot u'(X_u) \cdot B_1 + (1 - p) \cdot u'(X_d) \cdot B_1 - \lambda \cdot B_0 = 0 \\ &= E[u'(V_1) \cdot B_1] = \lambda B_0 \end{aligned} \tag{II}$$

$$\longrightarrow \quad \lambda = \frac{E[u'(V_1) \cdot B_1]}{B_0}$$

We assume that $B_0 = 1$ and $B_1 = (1 + r)$, we thus have:

$$\frac{1}{\lambda} = \frac{1}{(1 + r) \cdot E[u'(V_1)]}$$

→ Substitute II in I:

$$S_0 = \frac{1}{1 + r} \cdot \frac{E[u'(V_1) \cdot S_1]}{E[u'(V_1)]}$$

We define the variable Z , which represents the marginal utility of V_1 divided by the expected value of the marginal utility of V_1 :

$$Z = \frac{u'(V_1)}{E^P[u'(V_1)]}$$

Since V_1 can take two values, Z takes also two values:

$$Z_{\text{up}} = \frac{u'(X_{\text{up}})}{p \cdot u'(X_{\text{up}}) + (1 - p) \cdot u'(X_{\text{down}})} \quad Z_{\text{down}} = \frac{u'(X_{\text{down}})}{p \cdot u'(X_{\text{up}}) + (1 - p) \cdot u'(X_{\text{down}})}$$

Z has two properties:

1. Since we assume increasing utility ($u'(V_1) > 0$): $Z > 0 \quad \forall \mathbb{R}$
2. The expected value of Z is equal to 1:

$$E^P[Z] = E^P\left[\frac{u'(V_1)}{E^P[u'(V_1)]}\right] = \frac{E^P[u'(V_1)]}{E^P[u'(V_1)]} = 1$$

Therefore, we can rewrite equation iii as:

$$S_0 = \frac{1}{1 + r} \cdot E^P[Z \cdot S_1]$$

Where: $E^P[Z \cdot S_1] = p \cdot Z_{\text{up}} \cdot S_{0,u} + (1 - p) \cdot Z_{\text{down}} \cdot S_{0,d}$

$$\rightarrow S_0 = \frac{1}{1 + r} \cdot p \cdot Z_{\text{up}} \cdot S_{0,u} + (1 - p) \cdot Z_{\text{down}} \cdot S_{0,d}$$

By dividing both sides by S_0 , we get:

$$1 + r = p \cdot Z_{\text{up}} \cdot u + (1 - p) \cdot Z_{\text{down}} \cdot d$$

We define q and $(1 - q)$ as:

$$q = p \cdot Z_{\text{up}} = p \cdot \frac{u'(X_{\text{up}})}{E[u'(V_1)]} \quad \text{and} \quad 1 - q = (1 - p) \cdot Z_{\text{down}} = (1 - p) \cdot \frac{u'(X_{\text{down}})}{E[u'(V_1)]}$$

We thus have:

$$\longrightarrow \quad 1 + r = q \cdot u + (1 - q) \cdot d$$

$$1 + r = q(u - d) + d$$

$$\boxed{q = \frac{(1 + r) - d}{u - d}}$$

-i still adding text